

ConformalGuard: False-Positive-Controlled Action Gating for LM Classifiers

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Abstract

Language model classifiers are increasingly deployed for high-stakes decisions such as phishing screening, content moderation, and fraud triage, yet validation-set thresholds provide no finite-sample control of deployment-time false positives. We present ConformalGuard, a model-agnostic calibration layer that wraps a fixed scoring function with finite-sample control over the rate at which legitimate inputs trigger an intervention tier. Our contribution is empirical and operational, not a new conformal theorem: zero-shot LM API scores can be coarse enough (10–11 unique values here) to make temperature scaling and isotonic regression operationally identical to raw thresholding for low-FPR action gating. We compare these methods with conformal calibration for fine-tuned BERT, Claude Sonnet 4, and GPT-4o on public email corpora. For Claude, conformal calibration reduces FPR from 0.040 to 0.008 while recall falls from 0.938 to 0.875. For deterministic GPT-4o scores, the 1% operating point is too conservative to deploy (recall 0.564); three-run stochastic averaging raises recall to 0.732 at FPR 0.005. At 2% prevalence with BERT, conformal calibration removes 25 false-positive reviews per 10,000 emails (95% CI [10, 43]). The guarantee is legitimate-event-rate control under exchangeability; discrimination, workload effects, and robustness to shift remain empirical.

1 Introduction

Language models are increasingly used as classifiers for consequential decisions. An LLM screens emails as phishing or safe. A fine-tuned BERT flags toxic content. A GPT-based system triages fraud alerts. In each case, the deployment question is the same: given a score, when should the system act, defer, or escalate?

The standard approach is to tune a threshold on a validation set. This is a point estimate with no finite-sample coverage guarantee. It can overfit to the validation distribution, and the resulting false-positive rate at deployment may differ substantially from the target across scorer families.

The consequences at deployment are severe and asymmetric. Consider a security team processing 50,000 emails per day at 1% phishing prevalence. A classifier tuned for 90%

recall with a 1% false-positive rate sends 500 legitimate emails to analyst review daily. If the true rate is 1.5%, a half-point miscalibration invisible in balanced evaluation, that count rises to 750. At realistic prevalence, the false-positive rate controls analyst workload, not recall. Model accuracy gains are eroded if the decision layer leaks false positives.

Standard post-hoc calibration methods, temperature scaling (Guo et al., 2017) and isotonic regression (Zadrozny & Elkan, 2002), improve probabilistic calibration on average but provide no finite-sample guarantee on the false-positive rate at a given threshold. Conformal prediction (Vovk et al., 2005) offers exactly this: for any fixed scoring function and any target rate α , it produces p-values satisfying $\Pr(p(X) \leq \alpha \mid Y=0) \leq \alpha$ under exchangeability, without distributional assumptions.

This paper asks a direct question: does conformal calibration provide better deployment-time false-positive control for LM classifiers than standard calibration methods when a threshold must be chosen before deployment labels are available?

We answer this by evaluating fine-tuned BERT, Claude Sonnet 4, and GPT-4o on the same public email corpus under four calibration strategies. Zero-shot LMs produce coarsely discretized scores: Claude returns only 11 unique values and deterministic GPT-4o returns 10. Temperature scaling and isotonic regression produce the same action boundary as raw validation-set tuning. At $\alpha = 0.01$ on the in-distribution test split, the non-conformal alternatives miss the target by about 1.2 percentage points on average (mean absolute violation), with Claude reaching a 3.0-point overshoot, while conformal calibration stays within target. This does not imply improved discrimination or robustness under arbitrary shift; it provides a pre-deployment operating point under exchangeability.

Our contributions:

1. **A systematic deployment comparison.** We evaluate raw thresholds, temperature scaling, isotonic regression, and conformal prediction across three LM scorers. In our in-distribution experiments, conformal calibration provides the tightest pre-deployment FPR control.
2. **A score-granularity diagnosis.** Coarse zero-shot outputs make standard post-hoc calibration operationally inert for low-FPR action thresholds. Full-scale GPT-4o stochastic averaging shows that richer score grids improve the achievable recall-FPR trade-off.
3. **Conformal action gating.** We map calibrated p-values to ordered intervention tiers (auto-clear, analyst review, specialist escalation, autonomous blocking) with cumulative legitimate-event-rate bounds for each tier or any stricter tier.
4. **Operational impact at low prevalence.** With BERT at 2% phishing prevalence, conformal calibration reduces false positives from 83 to 58 per 10,000 emails, a reduction of 25 reviews (95% CI [10, 43]).

2 Background and related work

2.1 Post-hoc calibration of classifiers

Temperature scaling (Guo et al., 2017) learns a single scalar parameter T on a validation set, dividing logits by T before softmax. It improves expected calibration error on average but provides no guarantee that the FPR at any specific threshold is controlled. The learned temperature can overfit to the validation distribution.

Isotonic regression (Zadrozny & Elkan, 2002) fits a non-decreasing step function mapping scores to probabilities. It is non-parametric and handles arbitrary miscalibration shapes, but again provides no finite-sample FPR guarantee at a given operating point.

Both methods assume the input scores have sufficient granularity to support meaningful recalibration. When scores take only a handful of distinct values, as we find with zero-shot LLMs (Section 4), temperature scaling reduces to a monotone rescaling that cannot change

Capability	Temp. scaling	Isotonic	ConU	CADES	Ours
Finite-sample FPR guarantee	No	No	Yes	Yes	Yes
Validity on tied score grids	No	No	N/A	N/A	Yes; utility limited
Compared across LM architectures	No	No	No	No	Yes
Tiered action gating	No	No	No	No	Cumulative bound

Table 1: Positioning relative to existing work

the rank ordering and isotonic regression collapses to a lookup table with the same decision boundaries as the raw threshold.

LLM uncertainty estimation and confidence calibration span verbalized confidence, internal signals, consistency estimates, and ensemble methods (Xia et al., 2025; Liu et al., 2025). Calibrating verbalized probabilities is especially relevant when an API exposes JSON probabilities rather than token log probabilities (Wang et al., 2024a). ConformalGuard is complementary: it does not make confidence intrinsically calibrated, but asks whether a frozen score supports a pre-deployment threshold with legitimate-event FPR control.

2.2 Conformal prediction

Conformal prediction (Vovk et al., 2005; Angelopoulos & Bates, 2023) provides distribution-free, finite-sample guarantees under exchangeability. Let $C_0 = \{x_i : Y_i = 0\}_{i=1}^n$ contain only confirmed-legitimate examples, let $s_i = g(x_i)$ under a fixed scoring function g , and let $s(x) = g(x)$ for a new example. The conformal p-value is:

$$p(x) = \frac{1 + \sum_{i=1}^n \mathbf{1}[s_i \geq s(x)]}{n + 1} \quad (1)$$

Under exchangeability, this p-value provides a finite-sample bound on the false-positive rate at any chosen significance level α (stated formally as Proposition 1 in Section 3.2). This is not a probability of phishing. It is a rank-based conformity score that controls the rate at which legitimate examples fall below the threshold. The method does not require continuous scores or smooth calibration curves, but deterministic p-values inherit tied score blocks and therefore do not remove coarse-score plateaus.

Recent conformal work relevant to LMs includes ConU (Wang et al., 2024b), TECP (Xu & Lu, 2025), CADES (Zhang et al., 2025), and the CP-for-NLP survey (Campos et al., 2024). Bates et al. (2021) and Angelopoulos et al. (2024) extend conformal ideas to general risk control. Tibshirani et al. (2019) address covariate shift, while Romano et al. (2020) develop adaptive conformal classification and randomized prediction sets.

Our contribution is not a new conformal theorem. It is an empirical deployment study of standard conformal and post-hoc calibration applied to LM classifier scores. Unlike randomized conformal classification, our deployed action rule is deterministic. Conformal calibration does not improve the score ordering; it selects a low-FPR operating point from legitimate calibration data before deployment labels are observed.

2.3 Positioning

Table 1 summarizes our positioning. Our study isolates how coarse zero-shot LM score grids affect standard post-hoc calibration for low-FPR action gating across multiple scorer families.

3 Method

3.1 Overview

ConformalGuard wraps any fixed scoring function $g(x) \rightarrow [0,1]$ with a calibration layer that converts raw scores into calibrated p-values, then maps p-values to ordered action tiers.

Step 1: Scoring. A frozen scoring function g produces a suspiciousness score for each input. In our experiments, g is instantiated three ways: (a) fine-tuned BERT-base-uncased producing $P(\text{phishing})$, (b) Claude Sonnet 4 prompted to return a phishing probability, (c) GPT-4o prompted identically.

Step 2: Calibration. Given the legitimate-only set C_0 and scores $s_i = g(x_i)$ from the frozen scorer, we compute the conformal p-value in Equation 1 for a test example x , with $s(x) = g(x)$.

For comparison, we also calibrate each scorer with: (a) a raw threshold tuned on the validation set, (b) temperature scaling learned on the validation set, and (c) isotonic regression fitted on the validation set.

Step 3: Action gating. Calibrated scores map to intervention tiers:

- $p > 0.10$: auto-clear (no analyst involvement)
- $0.01 < p \leq 0.10$: analyst review
- $0.005 < p \leq 0.01$: specialist escalation
- $p \leq 0.005$: reversible autonomous blocking

The cumulative legitimate-event rate for each tier or any stricter tier is bounded by the corresponding α threshold under exchangeability (Proposition 1).

3.2 Conformal calibration

Proposition 1. If the legitimate calibration and test examples are exchangeable, and the score function g is fixed before calibration, then $\Pr(p(X) \leq \alpha \mid Y=0) \leq \alpha$ for any $\alpha \in [0,1]$.

Proof sketch. Under exchangeability and in the absence of ties, the test score’s rank among the $n+1$ scores (n calibration + 1 test) is uniform. With ties, the deterministic p-value based on the count of calibration scores greater than or equal to the test score assigns the least favorable rank within the tied block and is therefore conservative (super-uniform). Hence the probability that p is at most α is at most α ; see Vovk et al. (2005, Chapter 2) and Angelopoulos & Bates (2023, §2).

We do not interpret $p(x)$ as the probability that x is phishing. It is a rank-based conformity score relative to the legitimate calibration pool.

What conformal adds on discretized scores. Monotone post-hoc transformations cannot change the ranking or split a tied score block, so they often retain the validation-selected action boundary. Deterministic conformal p-values also do not create new discrimination or split ties. Their value is different: calibration ranks determine, before deployment, which existing score boundary satisfies the target legitimate-event rate under exchangeability. An oracle raw threshold chosen with test labels can match that boundary exactly. Randomized tie-breaking can split tied blocks, but sacrifices repeatable per-input tier assignments (Appendix A).

Robustness to distribution shift. If the total variation distance between calibration and test legitimate distributions is ϵ , then $\Pr(p(X) \leq \alpha \mid Y = 0) \leq \alpha + \epsilon$. Monitoring can detect a warning sign but does not restore validity under arbitrary shift. Persistent shift requires weighted, stratified, or adaptive conformal methods.

Calibration set selection bias. Confirmed-legitimate examples are not a random sample: they arise from auto-cleared events, analyst-reviewed events, or customer confirmations, each introducing selection bias. We mitigate through stratified sampling that matches the

Split	Total	Legitimate	Phishing/Spam	Positive rate
Train	26,202	11,937	14,265	54.4%
Val	3,742	1,705	2,037	54.4%
Cal	2,000	2,000	0	0% (legit. only)
Test	7,991	3,911	4,080	51.1%

Table 2: Corpus splits and class distribution (39,935 emails)

feature distribution between calibration and test sets, for example ensuring similar proportions of emails with full SMTP headers versus body-only text, since feature availability affects scorer behavior and exchangeability. Importance-weighted calibration (Tibshirani et al., 2019) could further address this gap.

Rolling calibration and model updates. Each recalibration produces a new conformal procedure. After any scorer update, freeze the updated scorer, collect a fresh legitimate calibration set, rescore it, recompute ranks, and only then deploy. The ordering is train $\rightarrow$ validate $\rightarrow$ freeze $g \rightarrow$ calibrate on disjoint legitimate data $\rightarrow$ deploy.

3.3 What the guarantee does and does not cover

Controlled by design: The rate at which legitimate events enter each action tier, under exchangeability.

Not guaranteed: Recall (detection rate), subgroup fairness, or workload reduction. These are supported empirically, not formally.

This is a deliberate scope restriction. A system that controls only legitimate-event rates is not a complete safety solution. But it addresses the deployment problem that causes the most operational pain: unnecessary analyst reviews of legitimate traffic.

4 Experiments

We evaluate four calibration strategies (raw threshold, temperature scaling, isotonic regression, and conformal prediction) across three scoring models (fine-tuned BERT, zero-shot Claude Sonnet 4, and zero-shot GPT-4o) on a unified public email phishing corpus.

4.1 Data

We assemble a public email corpus from four source families:

- **Nazario Phishing Corpus** (Nazario, 2005). Verified real phishing with full SMTP headers.
- **Phishing Pot** (rf-peixoto, 2022). 7,069 raw honeypot phishing emails with Outlook/Exchange headers (4,393 in the final leakage-cleaned splits).
- **SpamAssassin** (Apache SpamAssassin Project, 2003). Public ham and spam corpus with full headers.
- **Enron-Spam AUEB** (Metsis et al., 2006). Legitimate email baseline (body text only, no SMTP headers) and spam.

After excluding the EPVME synthetic dataset (identified as a protocol-vulnerability test suite containing XSS payloads rather than phishing emails; see supplementary audit report), filtering bodies below 50 characters, deduplicating by SHA-256 body hash, and removing 16 normalized-body duplicates from the lower-priority train/validation splits, the corpus contains 39,935 emails (Table 2). Test and calibration were unchanged by the leakage cleanup.

The positive class includes both true phishing (Nazario, Phishing Pot; 28.7% of positives) and spam (Enron spam, SpamAssassin spam; 71.3%), following the convention of prior

work on these corpora. The calibration set contains only confirmed-legitimate emails from multiple ham sources.

Data quality. A source-confusion classifier (TF-IDF + logistic regression, 5-fold CV) achieves 97.7% accuracy at distinguishing sources from text. This is expected for public email corpora where phishing, spam, and legitimate emails have systematically different vocabulary. The calibration comparison is valid regardless because all four methods operate on the same score distributions. Cross-split duplicate check confirmed zero leakage after cleanup. The full audit report is in supplementary material.

4.2 Scoring functions

We evaluate three LM classifiers, each producing a phishing suspiciousness score $s \in [0,1]$:

1. **BERT-base-uncased (fine-tuned).** Fine-tuned on the training split for 3 epochs ($lr=2e-5$, $batch=16$, $seed=42$). Produces 1,176 unique score values across the corpus, a near-continuous distribution.
2. **Claude Sonnet 4 (zero-shot).** Prompted with email subject and body to return a phishing probability. Produces **11 unique score values** (0.0, 0.05, 0.1, 0.2, 0.3, 0.7, 0.8, 0.85, 0.9, 0.95, 1.0), a coarsely discretized distribution.
3. **GPT-4o (zero-shot).** Prompted identically. Produces **10 unique score values** (0.0, 0.1, 0.2, 0.3, 0.7, 0.8, 0.9, 0.95, 0.99, 1.0), similarly coarse.

All three scoring functions are frozen before calibration. The sharp contrast in score granularity between BERT (1,176 levels) and the zero-shot LLMs (10–11 levels) is central to our results.

4.3 Calibration methods

For each scorer, we apply four calibration methods:

1. **Raw threshold.** Choose threshold on the validation set to achieve target $FPR \leq \alpha$. No post-hoc calibration.
2. **Temperature scaling.** Learn T on the validation set by minimizing negative log-likelihood. Threshold the calibrated scores at the validation quantile matching α .
3. **Isotonic regression.** Fit a non-decreasing step function on the validation set. Threshold at the validation quantile matching α .
4. **Conformal prediction.** Compute conformal p-values on a disjoint calibration set of 2,000 legitimate-only examples. Threshold at α .

Methods 1 to 3 set their threshold on the validation set and hope it transfers to the test set. Method 4 sets its threshold on a separate calibration set with a finite-sample guarantee.

4.4 Main result: calibration comparison across LMs

Table 3 reports 95% bootstrap confidence intervals from 1,000 resamples. $FPR\ violation = observed\ FPR - target\ \alpha$; negative means the method stays below the target, positive means it overshoots.

Finding 1: conformal stays within target at $\alpha = 0.01$ on the in-distribution split. Across all three scorers, conformal calibration produces observed FPR below target. Every alternative overshoots on at least one scorer. The average absolute violation is 0.5pp for conformal versus about 1.2pp for the alternatives. Section 5.1 shows this does not imply validity under arbitrary source shift.

Finding 2: Standard calibration methods are ineffective on LLM scores. For Claude Sonnet 4, temperature scaling and isotonic regression produce results identical to raw thresholds:

LM	Calibration	Recall	Obs. FPR	FPR viol.	Precision	F1
BERT	Raw threshold	0.996 ± 0.001	0.008	-0.002	0.992	0.994
BERT	Temp. scaling	0.996 ± 0.001	0.008	-0.002	0.992	0.994
BERT	Isotonic reg.	0.997 ± 0.001	0.013	+0.003	0.988	0.993
BERT	Conformal	0.995 ± 0.001	0.006	-0.004	0.994	0.995
Claude	Raw threshold	0.938 ± 0.004	0.040	+0.030	0.961	0.949
Claude	Temp. scaling	0.938 ± 0.004	0.040	+0.030	0.961	0.949
Claude	Isotonic reg.	0.938 ± 0.004	0.040	+0.030	0.961	0.949
Claude	Conformal	0.875 ± 0.005	0.008	-0.002	0.991	0.930
GPT-4o	Raw threshold	0.887 ± 0.005	0.015	+0.005	0.984	0.933
GPT-4o	Temp. scaling	0.887 ± 0.005	0.015	+0.005	0.984	0.933
GPT-4o	Isotonic reg.	0.887 ± 0.005	0.015	+0.005	0.984	0.933
GPT-4o	Conformal	0.564 ± 0.007	0.001	-0.009	0.999	0.721

Table 3: FPR control across three LMs and four calibration methods ($\alpha = 0.01$)

Runs	Unique	Conformal			Raw threshold	
		Recall	FPR	F1	Recall	FPR
1	33	0.285	0.0049	0.442	0.872	0.0355
3	105	0.732	0.0051	0.843	0.882	0.0194

Table 4: Full-scale GPT-4o stochastic averaging ($\alpha = 0.01$)

FPR 0.040 in all three cases (4x overshoot of the 0.01 target). The reason is mechanical. Claude returns only 11 distinct scores. Temperature scaling applies $s \rightarrow \text{sigmoid}(\text{logit}(s)/T)$, a monotone transform that preserves rank ordering. With only 11 input values, the transform maps to 11 output values, and the threshold falls between the same two adjacent scores regardless of T . Isotonic regression faces the same limitation. The three methods are operationally identical on discretized inputs.

Finding 3: FPR control exposes a real utility trade-off. Claude FPR falls from 0.040 to 0.008 while recall falls from 0.938 to 0.875. For deterministic GPT-4o, conformal reaches FPR 0.001 but recall falls from 0.887 to 0.564. This operating point is not deployable under a typical recall floor. Conformal selects a valid pre-deployment boundary; it does not improve classifier discrimination or overcome a poorly resolved score grid.

Finding 4: BERT is the control case. With 1,176 unique values, raw thresholding and temperature scaling behave similarly to conformal, while isotonic regression slightly overshoots (FPR 0.013). The small raw-conformal gap (0.008 vs. 0.006) supports score granularity, rather than the dataset alone, as the source of the larger LM gaps.

4.5 Full-scale GPT-4o stochastic averaging

The deterministic run is a fixed-scorer evaluation, but its ten-level output is unusually coarse. We therefore score the complete calibration, validation, and test splits at nonzero temperature and compare one stochastic draw with the mean of three independent draws, freezing each resulting pipeline before calibration (Table 4). Averaging increases unique scores from 33 to 105 and improves conformal recall from 0.285 to 0.732 at nearly unchanged FPR (0.0049 to 0.0051). The raw validation-selected threshold also improves, but still exceeds the 0.01 target (FPR 0.0194). Thus richer scoring improves the recall-FPR frontier; conformal calibration supplies threshold validity rather than the discrimination gain.

4.6 Low-prevalence evaluation

At realistic inbox prevalence, false positives dominate analyst workload. We simulate 2% phishing prevalence using BERT by keeping all 3,911 legitimate test emails and subsampling positives. We average prevalence-weighted precision over 50 repetitions; recall is measured

Calibration	Recall	FPR	Precision	FP per 10K emails (95% CI)
Raw threshold	0.996	0.008	0.707	83 [55, 113]
Temp. scaling	0.996	0.008	0.707	83 [55, 113]
Conformal	0.995	0.006	0.776	58 [35, 83]

Table 5: Low-prevalence evaluation (2% phishing, BERT scorer)

Calibration n	FPR (mean $\pm$ std)	Recall (mean $\pm$ std)
100	0.006 $\pm$ 0.006	0.988 $\pm$ 0.012
500	0.006 $\pm$ 0.003	0.994 $\pm$ 0.002
1,000	0.006 $\pm$ 0.002	0.995 $\pm$ 0.001
2,000 (full)	0.006 $\pm$ 0.000	0.995 $\pm$ 0.000

Table 6: FPR and recall vs. calibration set size (BERT, $\alpha = 0.01$)

on the full test positives, and false-positive intervals use paired bootstrap resampling of legitimate test emails (Table 5).

At 2% prevalence, conformal calibration reduces false positives from 83 to 58 per 10,000 emails: 25 fewer reviews (paired-bootstrap 95% CI [10, 43]). Precision rises from 0.707 to 0.776, while recall is statistically indistinguishable. Isotonic regression is omitted from this workload table because it already has the larger BERT FPR violation in Table 3; including it would not affect the raw-versus-conformal workload comparison.

4.7 Calibration set sensitivity

How many legitimate examples are needed for effective conformal calibration? We subsample the calibration set for the BERT scorer at $\alpha = 0.01$ (Table 6).

Mean FPR remains below target at $n = 100$, but the high variance means individual calibration draws can exceed it. Recall stabilizes at $n \geq 500$. We recommend at least 1,000 legitimate examples and full human review during cold start.

5 Discussion and limitations

Calibration is not discrimination. An oracle raw threshold selected with test labels can reproduce the deterministic conformal boundary because both act on the same score ordering. Conformal’s advantage is pre-deployment selection using legitimate calibration data, not improved ranking. This distinction matters for GPT-4o: deterministic conformal recall drops from 0.887 to 0.564, so we would reject that scorer for 1% FPR gating. Three-run averaging improves the frontier, but adds inference cost and requires the averaging procedure itself to be frozen before calibration.

Randomization is a different interface. Randomized conformal tie-breaking raises GPT-4o recall to 0.800 with mean FPR 0.011, a slight finite-test-split fluctuation above $\alpha = 0.01$, but 17.2% of test decisions depend on the random variate and seed-to-seed disagreement is 7.0% (Appendix A). Deterministic tiers preserve repeatability for analyst-facing actions; randomized methods remain a useful population-level baseline rather than a free solution to coarse scores.

5.1 Exchangeability and source shift

The guarantee assumes exchangeable legitimate calibration and test examples. Table 7 isolates this limitation by calibrating and testing on different legitimate source families; within-source baselines appear in Appendix A. BERT calibrated on Enron ham and tested on SpamAssassin ham reaches FPR 0.021 at $\alpha = 0.01$; GPT-4o reaches 0.017 in one reverse direction. Monitoring can trigger fallback or recalibration, but cannot restore validity under

Scorer	Cal. source	Test source	n_{test}	FPR
BERT	Enron	SpamAssassin	801	0.021
BERT	SpamAssassin	Enron	3,110	0.003
Claude	Enron	SpamAssassin	801	0.006
Claude	SpamAssassin	Enron	3,110	0.008
GPT-4o	Enron	SpamAssassin	801	0.000
GPT-4o	SpamAssassin	Enron	3,110	0.017

Table 7: All cross-source legitimate-shift directions ($\alpha = 0.01$)

arbitrary shift. Persistent shift calls for stratified, weighted, or adaptive conformal methods (Tibshirani et al., 2019; Barber et al., 2023).

Dataset and model scope. A source-confusion classifier reaches 97.7% accuracy, so discrimination may partly exploit corpus artifacts; all calibration methods inherit that scorer. The positive class combines phishing (28.7%) and spam (71.3%), making the task malicious/unwanted-email detection rather than pure phishing. Claude and GPT-4o are zero-shot whereas BERT is fine-tuned, and our results do not establish how fine-tuned open LMs behave.

What remains outside the guarantee. We do not guarantee recall, workload reduction, adversarial robustness, or subgroup fairness. Attacks on positive inputs change detection rather than the legitimate-event FPR, while calibration-pool poisoning and legitimate-distribution attacks directly threaten exchangeability. Multiclass prediction sets, regression coverage, conditional fairness control, stronger attacks, and fine-tuned open-LM evaluations are future work.

6 Conclusion

We showed that conformal calibration can select tighter pre-deployment false-positive operating points than validation-selected temperature scaling and isotonic regression for LM classifiers with coarse score grids. This is a standard conformal guarantee used as an operational action-gating interface, not a new theorem or a claim of improved discrimination.

For Claude Sonnet 4, FPR falls from 0.040 to 0.008 at a 6.3-point recall cost. Deterministic GPT-4o exposes an unacceptable recall loss, while three-run averaging recovers recall to 0.732 at FPR 0.005. At 2% prevalence with BERT, calibration avoids 25 false-positive reviews per 10,000 emails (95% CI [10, 43]). These are operating-point choices whose validity depends on exchangeability and whose utility must be evaluated separately.

The practical rule is simple: freeze the complete scorer, calibrate on fresh legitimate examples, gate actions by p-value tier, and reject the scorer when the target FPR implies unacceptable recall. After any model or prompting change, rescore and recalibrate before deployment.

Reproducibility statement

The evaluation uses exclusively public datasets: Nazario Phishing Corpus (Nazario, 2005), Phishing Pot (rf-peixoto, 2022), SpamAssassin (Apache SpamAssassin Project, 2003), and Enron-Spam AUEB (Metsis et al., 2006). The EPVME dataset was excluded after a quality audit identified it as a protocol-vulnerability test suite rather than a phishing corpus; the audit report is in supplementary material. The public camera-ready artifact is available at <https://github.com/bishnubista/conformalguard-colm-2026>. The package contains pinned data-download and parsing scripts, SHA-256 provenance, the leakage-cleaned split manifest, exact training and prompting configurations, calibration code, score-only records, and reported outputs. The public export omits raw messages and parsed email bodies.

LLM use disclosure

LLMs are used in two capacities: (1) as classification systems under study (fine-tuned BERT, Claude Sonnet 4, GPT-4o) and (2) for writing assistance. All reported numbers are verified against bundled output files. The evaluation prompts are included in the repository.

Ethics statement

All evaluation data comes from publicly released research corpora: Nazario Phishing Corpus (Nazario, 2005), Phishing Pot (rf-peixoto, 2022), SpamAssassin (Apache SpamAssassin Project, 2003), and Enron-Spam AUEB (Metsis et al., 2006). No customer data was used. Because these corpora can contain real messages and personal information, the public artifact omits raw messages and parsed bodies and instead provides pinned retrieval scripts, hashes, pseudonymous score records, and aggregate outputs. The EPVME synthetic dataset was excluded on data-quality grounds. Production deployment would require data minimization, regulatory compliance, and fairness auditing for disparate impact.

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A Calibration sensitivity and robustness analyses

A.1 Class-balanced calibration ablation

Adding positives when fitting temperature scaling and isotonic regression does not change the Claude or GPT-4o decisions at $\alpha = 0.01$ (Table 8). Score granularity, not class composition of the fit set, remains the bottleneck.

A.2 Sensitivity to the target false-positive rate

Table 9 shows smooth movement for BERT but plateaus for coarse LM scores. Claude is unchanged between $\alpha = 0.01$ and 0.05; GPT-4o is unchanged from 0.001 through 0.01.

Scorer	Method	Recall	FPR	Precision	F1
BERT	Temp.	0.9963	0.0084	0.9919	0.9941
BERT	Isotonic	0.9973	0.0130	0.9876	0.9924
Claude	Temp.	0.9377	0.0399	0.9608	0.9491
Claude	Isotonic	0.9377	0.0399	0.9608	0.9491
GPT-4o	Temp.	0.8870	0.0153	0.9837	0.9329
GPT-4o	Isotonic	0.8870	0.0153	0.9837	0.9329

Table 8: Balanced-validation calibration ablation

Scorer	$\alpha = .001$		.005		.01		.05		.10	
	Rec.	FPR	Rec.	FPR	Rec.	FPR	Rec.	FPR	Rec.	FPR
BERT	.968	.001	.992	.003	.995	.006	.999	.052	1.000	.097
Claude	.040	.000	.729	.002	.875	.008	.875	.008	.970	.060
GPT-4o	.564	.001	.564	.001	.564	.001	.887	.015	.953	.067

Table 9: Deterministic conformal threshold sensitivity

A.3 Within-source false-positive baselines

For Enron ($n = 3,110$) and SpamAssassin ($n = 801$), respectively, the matched-source FPRs are BERT 0.006/0.011, Claude 0.008/0.006, and GPT-4o 0.001/0.008. The BERT SpamAssassin result is a one-count-scale finite-test-split overshoot at $\alpha = 0.01$, not evidence that matched source alone guarantees exact empirical control.

A.4 Randomized conformal tie-breaking

For G calibration scores greater than a test score and E equal to it, we use $p = (G + U(E + 1)) / (n + 1)$ with $U \sim \text{Unif}(0, 1)$ and report 100-seed means (Table 10). Randomization improves GPT-4o recall but changes the per-input decision contract; its mean test FPR of 0.011 is a finite-test-split fluctuation around $\alpha = 0.01$.

Scorer	Recall	FPR	F1	Depends on U	Seed disagreement
BERT	0.995	0.006	0.995	0.0%	0.0%
Claude	0.881	0.010	0.932	4.7%	0.7%
GPT-4o	0.800	0.011	0.883	17.2%	7.0%

Table 10: Randomized conformal tie-breaking at $\alpha = 0.01$